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Disturbance detection in landsat time series is influenced by tree mortality agent and severity, not by prior disturbance

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ABSTRACT

Landsat time series (LTS) and associated change detection algorithms are useful for monitoring the effects of global change on Earth's ecosystems. Because LTS algorithms can be easily applied across broad areas, they are commonly used to map changes in forest structure due to wildfire, insect attack, and other important drivers of tree mortality. But factors such as initial forest density, tree mortality agent, and disturbance severity (i.e., percent tree mortality) influence patterns of surface reflectance and may influence the accuracy of LTS algorithms. And while LTS algorithms are widely used in areas with a history of multiple disturbance events during

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